UM-1: the coupled field equations

dirac16complex project (with Claude Opus 5.5)

Contents

1. Notation **2**

2. Gravity: the background (order ϵ^0) **2**

3. Sources, species and the four-dimensional observer **3**

4. Gravity at order ϵ: scalar perturbations **3**

5. The x_0 mass mechanism (the author's idea, as equations) **4**

6. dirac16complex: the Kohn-Sham equations coupled to all fields **4**

7. The Standard-Model equations (ASSUMED structure, standard form in the curved eight-dimensional background) **5**

8. Gauge invariance and the W **5**

9. The coupled system **5**

10. Checks (`field_equations/reports/field-equations-checks.json`, 13 of 13 PASS) **6**

Version 1 of 2026-10-17, for the model of `MODEL.md`. Derived by `field_equations/derive_field_equations.py` (sympy; 13 checks, all PASS, 2.7 s; outputs `field_equations/field-equations.json` with every equation as a sympy string and `field_equations/reports/field-equations-checks.json`). Labels as in `MODEL.md`: DERIVED (computed here), TAKEN (from the Revision record), ASSUMED, OPEN. Run: `python Revision/universe_model/field_equations/derive_field_equations.py`.

1. Notation

The author's coordinates $x_0, \dots, x_7$ (x_0 hidden, period $\ell_0 = \pi/(6H)$; x_4 the time). The hidden coordinate $y = \ln|\sin z|/(6H) \in [-L, 0]$ on each half of the cell ($dy = \cot z\, dx_0$, fold $y = 0$, regular tip $y = -L$). The background functions $a_4(x_4)$, $b_4(x_4)$; dots or primes on them are d/dx_4; $u = \dot{b}_4/b_4$, $v = \dot{u}$, $k = e^{2a_4}/b_4$; the block scale factor $f = e^{-a_4} b_4^{1/2}$ and its rate $h_4 = \dot{f}/f = u/2 - \dot{a}_4$.

In the y chart the background metric is, for EVERY real z (DERIVED, check A3 and the chart remark of the script),

$$\bar{g} = \mathrm{diag}(e^{-2a_4} b_4,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4} b_4 e^{2Hy},\ -e^{-2a_4} b_4 e^{2Hy},\ -e^{-2a_4} b_4 e^{2Hy}),$$

because $g_{00} dx_0^2 = e^{-2a_4} b_4\, dy^2$ and $(\sin^2 z)^{1/6} = e^{2Hy}$; the measure is $\sqrt{|g|}\, dx_0 = e^{-a_4} b_4^2 e^{3Hy}\, dy$ (check A3). The mixed tensor components are the same in both charts.

2. Gravity: the background (order ϵ^0)

- The Einstein tensor, recomputed here from the metric with its own Christoffel symbols, equals the record (`Revision/field_equations_a4/a4-equations.json`, key einstein; check A1):

$$G^{x_4}{}_{x_4} = 3\dot{a}_4^2 + 6H^2 k - \tfrac{3}{2}u^2, \quad G^{x_1}{}_{x_1} = -5\dot{a}_4^2 + 2\ddot{a}_4 + 6\dot{a}_4 u - \tfrac{5}{2}u^2 - 2v + 6H^2 k, \quad G^{x_0}{}_{x_0} = G^{x_5}{}_{x_5} = -3\dot{a}_4^2 + \tfrac{3}{2}\dot{a}_4 u - \tfrac{3}{2}u^2 - \tfrac{3}{2}v + 3H^2 k$$

and every off-diagonal component vanishes.

- The Einstein-Lovelock equations $\sum_k \alpha_k E_{(k)}{}^M{}_N + \Lambda \delta^M_N = \kappa T^M{}_N$ have three independent components, x_4 (a first-order CONSTRAINT), $x_1 = x_2 = x_3$ and $x_0 = x_5 = x_6 = x_7$ (the latter two linear in $(\ddot{a}_4, v)$); their full forms with the author's weights are TAKEN from the record (generalSource: constraint_x4, space_x1_eq_x2_eq_x3, hidden_and_extraTimes_x0_eq_x5_eq_x6_eq_x7), and `SOLUTION.md` integrates them.
- With $u = 2(h_4 + \dot{a}_4)$ the Einstein constraint reads (check A2)

$$3\dot{a}_4^2 + 12\, \dot{a}_4 h_4 + 6h_4^2 - 6H^2 k = \kappa\rho + \Lambda.$$

For a static block ($h_4 = 0$) it is the Friedmann equation with the effective cosmological constant $\Lambda_{eff} = \Lambda + 6H^2 k$.

- The source must be diagonal, independent of x_0 and have $p_0 = p_t$ (TAKEN: hiddenEqualsExtraTimes, offDiagonal, x0_dependence). Sources that violate this at a small level (the

interacting Kohn-Sham states of Ψ, the localised Standard-Model fermions) drive the x_0-dependent and the block-anisotropic metric corrections of order ϵ and higher (OPEN beyond the zero-mode averages used in `SOLUTION.md`).

3. Sources, species and the four-dimensional observer

- Every species i (photons, neutrinos, electrons, baryons, the Ψ components) contributes $(\rho_i, p_{3i}, p_{0i} = p_{ti})$; each obeys (check B1)

$$\dot{\rho}_i = -3\dot{a}_4(\rho_i + p_{3i}) - 4h_4(\rho_i + p_{0i}) + Q_i, \qquad \sum_i Q_i = 0,$$

with Q_i the energy exchanged through the portal interactions. The four-dimensional density $\rho_{4i} = V_\perp \rho_i$, $V_\perp \propto f^4$, obeys $\dot{\rho}_{4i} = -3\dot{a}_4(\rho_{4i} + p_{3,4i}) - 4h_4 p_{0,4i} + Q_{4i}$: the standard law for every species without hidden or extra-time pressure.

- The observer's Friedmann equation (Einstein part): $3\dot{a}_4^2 = \rho_4/M_{Pl}^2 + \Lambda_{eff} - 12\dot{a}_4 h_4 - 6h_4^2$ with $M_{Pl}^2 = V_\perp/\kappa = 2f^4 v_t(1 - e^{-3HL})/(3H\kappa)$.
- The drifts (check B2): $\dot{k} = -2kh_4$ and $(f^4)^{\cdot} = 4h_4 f^4$. A deflating block changes Λ_{eff}, the Planck mass and every x_0-generated mass ($\propto 1/f$): the observations need $|h_4| \ll \dot{a}_4$ after the primordial era.
- The block is NOT free: the x_0 component is its equation of motion. For a static block and Einstein gravity it reduces to $6\dot{a}_4^2 + 3\ddot{a}_4 = 3H^2 k + \Lambda - \kappa p_0$; matter with $p_0 = 0$ in an expanding universe cannot keep the left-hand side constant, so the Standard-Model matter DRIVES the block (the radion problem of every compactification). `SOLUTION.md` integrates the full system.

4. Gravity at order ϵ: scalar perturbations

The metric $g = \bar{g} + \epsilon\,\delta g$ with $\delta g_{44} = -2\Phi$, $\delta g_{ii} = -2\Psi_N e^{2a_4}$ $(i = 1, 2, 3)$, $\delta g_{00} = 2\sigma\bar{g}_{00}$, $\delta g_{tt} = 2\sigma\bar{g}_{tt}$ $(t = 5, 6, 7)$, with Φ, Ψ_N, σ functions of (x_4, x_1) (zero modes of x_0; one Fourier direction suffices for the scalar sector). The linearised Einstein tensor, computed from the metric (check C1; free of x_0; $\delta G^{x_4}{}_{x_0} = \delta G^{x_1}{}_{x_0} = 0$; $\delta G^{x_0}{}_{x_0} = \delta G^{x_5}{}_{x_5}$), with $\nabla^2 = e^{-2a_4}\partial_1^2$:

$$\delta G^4{}_4 = 3(u^2 - 2\dot{a}_4^2)\Phi + 3(2u - 2\dot{a}_4)\dot{\Psi}_N - 6u\dot{\sigma} - 12H^2 k\sigma - 2\nabla^2\Psi_N + 4\nabla^2\sigma$$

$$\delta G^4{}_1 = \partial_1\big[2(\dot{a}_4 - u)\Phi - 2\dot{\Psi}_N + (2u - 8\dot{a}_4)\sigma + 4\dot{\sigma}\big]$$

$$\delta G^1{}_1 - \delta G^2{}_2 = \nabla^2(\Psi_N - \Phi - 4\sigma)$$

The full $\delta G^1{}_1$ and $\delta G^0{}_0 = \delta G^5{}_5$ are in `field-equations.json` (key order_epsilon_scalar). The equations are $\delta G^M{}_N = \kappa\,\delta T^M{}_N$ ($\delta T^4{}_4 = -\delta\rho$, $\delta T^4{}_1$ the momentum density, $\delta T^1{}_1 = \delta p_3$, $\delta T^0{}_0 = \delta T^5{}_5 = \delta p_0$).

- Static block, no radion (check C2): $\delta G^4{}_4 = 6\dot{a}_4(\dot{\Psi}_N + \dot{a}_4\Phi) - 2\nabla^2\Psi_N$ and $\delta G^4{}_1 = -2\partial_1(\dot{\Psi}_N + \dot{a}_4\Phi)$: the Newtonian-gauge equations of four-dimensional general relativity with $H_{obs} = \dot{a}_4$. The CMB and BAO physics of `SOLUTION.md` uses this limit.
- Without anisotropic stress the slip is $\Psi_N = \Phi + 4\sigma$ (DERIVED): a radion perturbation is observable as a gravitational slip.
- The Gauss-Bonnet and cubic Lovelock contributions at order ϵ, and the vector and tensor sectors, are OPEN in version 1; at late times their background coefficients enter through $\alpha_2\dot{a}_4^2$, $\alpha_2 H^2 k$ and the like (the record's components), which `SOLUTION.md` evaluates.

5. The x_0 mass mechanism (the author's idea, as equations)

- Gauge fields (check D1): a four-dimensional component $A_\mu(x)u(y)$ (gauge $A_{x_0} = 0$) has, on each half of the cell, the action $\int_{-L}^{0} dy\, e^{3Hy}\big[u'^2/f^2 - m^2u^2\big] + c_{tip}u(-L)^2$, hence

$$-\frac{1}{f^2}e^{-3Hy}\big(e^{3Hy}u'\big)' = m^2u, \qquad u'(0) = 0, \qquad \frac{e^{-3HL}}{f^2}u'(-L) = c_{tip}\,u(-L).$$

- Massless zero modes (check D2): $c_{tip} = 0$ admits $u = \text{const}$, $m = 0$: the photon, the gluons and (for g_{MN}) the graviton. To first order in c_{tip}: $m^2 = c_{tip}/\int_{-L}^{0} e^{3Hy}dy = 3Hc_{tip}/(1 - e^{-3HL})$; the tip sigma field gives $c_{tip}(W) = \hat{g}^2 v_\Sigma^2 w_{tip}/4$, $c_{tip}(Z) = (\hat{g}^2 + \hat{g}'^2)v_\Sigma^2 w_{tip}/4$, $c_{tip}(\gamma) = 0$, so $m_Z/m_W = 1/\cos\theta_W$ ($\rho = 1$) at this order; the bending of the profiles beyond it is computed in `SOLUTION.md`.
- Fermions (check D4): the block $h = -(i/f)\sigma_1\partial_y + \mu\sigma_2$ annihilates $(e^{f\mu y}, 0)$ (fold $b(0) = 0$, tip $b(-L) = 0$; left-handed) and $(0, e^{-f\mu y})$ (fold $a(0) = 0$, tip $a(-L) = 0$; right-handed, theorem T3). Their normalised tip values are $\zeta(\mu)^2 = 2f\mu\, e^{-2f\mu L}/(1 - e^{-2f\mu L})$, and the tip coupling gives $m_f = \lambda_f\,\zeta(\mu_L)\,\zeta(\mu_R)$: exponentially small masses for $f\mu L \gg 1$.
- The uncertainty relation (check D3): the mass squared of a bosonic mode is its mean squared hidden momentum (Rayleigh quotient), so $m \geq \hbar/(2c\,\Delta s)$ for a profile of proper width $\Delta s = f\Delta y$, with equality for a Gaussian. A CHIRAL zero mode is the exception: its hidden momentum is cancelled by the bulk mass ($h\chi = 0$), it is massless however localised it is, and its mass comes from its value at the tip (DERIVED; `SOLUTION.md` section 4 quantifies it).

6. dirac16complex: the Kohn-Sham equations coupled to all fields

The good-sector reduction of the record (TAKEN, `Revision/kohn_sham/ks-theory.json`) with the portal mean fields (DERIVED from P1, P2, P5 of `MODEL.md`):

$$h_j = j\big[-(i/f)\sigma_1\partial_y + M_{eff}(y)\sigma_2 + \kappa k\sigma_3\big] + v_v(y), \qquad \kappa = e^{-a_4},$$

$$M_{eff} = m + \tfrac{15}{16}\lambda S(y) + g_S\sum_f s_f\,\delta_{tip}(y) + g_D\sum_f m_f s_f(y), \qquad v_v = -\tfrac{1}{16}\lambda n(y) + g_V\sum_f q_f n_f(y),$$

with $n_f(y) = n_f|\chi_f(y)|^2$ the Standard-Model fermion densities in their x_0 zero modes, s_f their scalar densities, $g_V = c_V/\Lambda_D^6$, $g_S = c_S/\Lambda_D^6$, $g_D = c_D/\Lambda_D^7$. P3 and P4 have no mean field in a neutral, unpolarised background with $\Sigma = 1$ and act through scattering. The back-reaction: the Standard-Model masses shift by $g_S S_\Psi(-L)\zeta_L\zeta_R + g_D m_f\langle S_\Psi\rangle_f$, the vector potentials by $g_V q_f n_\Psi(y)$, and the gravitational source is the total $T^M{}_N$ (the portal energy counted once). The Kohn-Sham energy-momentum of Ψ ($\rho = \sum wgf\varepsilon n_o - e_{int}$, p_3, $p_t = e_{int}$, p_0; record) enters section 3 as one species; its $p_0 \neq p_t$ in interacting states is a source of order-ϵ block anisotropy.

7. The Standard-Model equations (ASSUMED structure, standard form in the curved eight-dimensional background)

- Fermions: $\gamma^M D_M\psi - \mu_\psi\psi - (\text{tip terms})\,\delta_{tip} - (\text{portal terms}) = 0$ with $D_M = \partial_M + \Omega_M - i\hat{g}_s G_M^\alpha T^\alpha - i\hat{g}W_M^a\tau^a/2 - i\hat{g}'YB_M$.
- Gauge fields: $\frac{1}{\sqrt{|g|}}\partial_M(\sqrt{|g|}F^{MN}) + [\,A_M, F^{MN}] = J^N_{SM} + J^N_{portal} + J^N_{tip}\delta_{tip}$, with the portal current of P3, $J^N \ni \frac{\mu_D}{\Lambda_D^3}\nabla_M(\bar{\Psi}S^{MN}\Psi)$ for B, and the sigma-field current at the tip.
- The tip sigma field: $D_\mu(\Sigma^\dagger D^\mu\Sigma) = 0$ plus the fermion and portal terms (its equation of motion; unitary gauge $\Sigma = 1$).

8. Gauge invariance and the W

- The tip Yukawa terms $\bar{L}\Sigma\Lambda_\ell\ell_R$, $\bar{Q}\Sigma\Lambda_q q_R$ and the operator $\mathrm{tr}(\Sigma^\dagger D\Sigma\tau^3)$ are invariant under $SU(2)_L \times U(1)_Y$ to 10^{-15} in an explicit representation with 20 random group elements (check E1); P1, P3, P5 are products of the singlet $\bar{\Psi}\Gamma\Psi$ with invariant Standard-Model operators.
- A Ψ bilinear is neutral, so a vertex with one $W^\pm$ needs a charged Standard-Model pair: the W enters the portal only through the covariant derivative of P5 and through the gauge self-couplings (check E2).

9. The coupled system

equation	**unknowns**	**couples to**
Einstein-Lovelock background (3 components)	a_4, b_4	every species through $\rho, p_3, p_0 = p_t$
species conservation	ρ_i	a_4, h_4, the portal exchanges Q_i
order-ϵ scalar equations	Φ, Ψ_N, σ	$\delta\rho_i$, velocities, δp_0
x_0 eigenproblems	masses, profiles	f (hence h_4), H, L, c_{tip}, μ_ψ, λ_f
Kohn-Sham equations of Ψ	levels, n, S of Ψ	the Standard-Model densities (portals), a_4, f

Standard-Model field equations	ψ, A, Σ	Ψ (portals), the metric

OPEN in version 1: the Lovelock terms and the vector and tensor sectors at order ϵ; the x_0-dependent metric response to the localised sources; the derivation of the portal and tip coefficients.

10. Checks (`field_equations/reports/field-equations-checks.json`, 13 of 13 PASS)

A1 background Einstein tensor = record; A2 constraint in h_4; A3 the measure in y; B1 four-dimensional conservation; B2 the drifts of k and the Planck mass; C1 linear scalar perturbations; C2 the static-block limit is four-dimensional general relativity, slip $\Psi_N = \Phi + 4\sigma$; D1 the gauge x_0 operator and its boundary conditions; D2 massless zero modes, the leading W, Z masses, $\rho = 1$; D3 the uncertainty bound; D4 the fermion zero modes and tip values; E1 gauge invariance; E2 the W and charge conservation.